

STANDARD OPERATING PROCEDURE

SOP NO.: MHPL/VL/005-02	RH AND TEMPERATURE AREA STUDY	Date Prepared.: - 18/09/2025
Supersedes.: MHPL/VL/005-01		Effective Date.: - 10/10/2025
		Next review Date.: - 09/10/2027
		Page.: 1 of 3

1.0 PURPOSE

To define a standardized procedure for performing Temperature and Relative Humidity (RH) studies as part of HVAC system qualification and environmental validation services. This study ensures that controlled areas such as cleanrooms, and storage zones maintain required environmental conditions in compliance with regulatory standards WHO and helps identify critical temperature or humidity variations.

2.0 SCOPE

This SOP applies to all temperature- and RH-controlled areas such as warehouses, cold rooms, cleanrooms, and stability chambers used for the storage or processing of pharmaceutical materials/products.

3.0 RESPONSIBILITIES

Validation Engineer /service Engineer/ Service Technician.

4.0 DEFINITION**Temperature Mapping:**

A process to determine temperature distribution within a space over time.

RH Mapping:

A process to determine relative humidity variation within a space over time.

Data Logger:

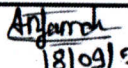
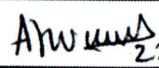
A device used to record temperature and RH at defined intervals.

Worst-case Location:

The location with maximum deviation from the set limits.

5.0 PROCEDURE

- 5.1.** Ensure that the area and equipment qualification has been completed and that the equipment is in proper running condition.

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- 5.2. First, measure the length, width, and height of the area or equipment to calculate its volume in cubic meters. Determine the minimum number of data loggers required for mapping using the formula $\sqrt{N}$, where N is the volume (in m^3). Additional EDLMs may be used if necessary to ensure comprehensive coverage.
- 5.3. Classify the area and divide it into grid cells based on its dimensions. For areas with a height of 3.6 meters or less, further divide each grid vertically into top, middle, and bottom sections. Arrange the data loggers in a zigzag pattern across the grids to ensure uniform spatial coverage.
- 5.4. If the ceiling height exceeds 3.6 meters, EDLMs should be arranged vertically at the bottom, middle, and top levels. For example, in a storage area with a 6-metre height, EDLMs can be placed at 0.3 m, 1.8 m, 3.6 m, and 5.4 m within each grid location. (See in Figure No;01).

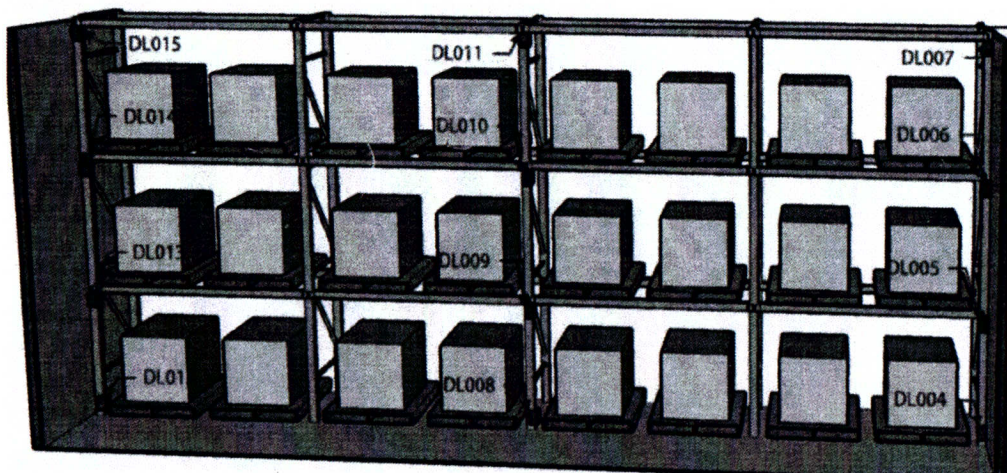
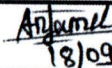
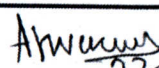


Fig No;01: Typical location of data loggers in a pallet racking storage area

- 5.5. Use only calibrated data loggers for conducting temperature and relative humidity (RH) mapping to ensure accuracy and reliability of the recorded data.
- 5.6. Set the data logger parameters according to client requirements, including logging interval, duration, temperature/RH limits, and alarm settings.

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- 5.7. After completing the setup, place the data loggers at their specified locations. Once positioned, they will automatically begin recording temperature and RH data as per the configured parameters.
- 5.8. After completing the mapping study, remove the data loggers and download the recorded data in compliance with 21 CFR Part 11 requirements, as per customer or regulatory specifications.
- 5.9. After the mapping is completed, evaluate the results for pass or fail status and submit the final report to the customer.

6.0 ANNEXURE

NA

7.0 REFERENCE

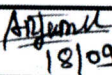
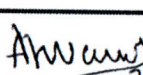
WHO Technical Report Series, No. 961, 2011

8.0 ABBREVIATION

RH: Relative humidity.
AHU: Air handling unit.
EDLMs: Electronic data logging monitor system.

9.0 REVISION HISTORY

Revision No.	Effective Date	Reasons For Revision
00	21/10/2021	First Issue.
01	15/10/2023	Periodic review and brief description of Temperature & Relative Humidity mapping is to be carried out by using Data logger.
02	10/10/2025	SOP elaborated as per standard regulatory requirements and revised during periodic review.

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